HONEY-B2024 LTD

INSTALLATION GUIDE

TOOD-OS Installation and Agent Execution Guide

How an AI coding agent installs TOOD-OS in VS Code, and how the development team tests it before use

CONTROLLED DOCUMENT

> Open the repository in VS Code, run one bootstrap script, and an agent works through ten tasks. The development team then runs sixteen tests before anyone uses the system.

VERSION	DATE
1.1	9 October 2026
OWNER	**STATUS**
Honey-B2024 Ltd	Controlled document
COMPANION TO	
Master Blueprint v1.1	

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Contents

1 The pack 3
2 What gets installed 3
3 Prerequisites 4
4 Folder structure 5
5 Install in VS Code 5
6 Agent execution protocol 6
7 Installation tasks 7
8 What success looks like 8
9 Development team test plan 8
10 After installation: what remains to build 9
11 Troubleshooting 10
12 Hand-over checklist 11
Contact 11

> ***In short:*** *open the repository in VS Code, run one bootstrap script, and an agent works through ten tasks (T00 to T09). The install is finished when* `verify_install.py` *prints* `16/16 checks passed` *and the test suite shows* `18 passed`*. The development team then runs sixteen tests (D01 to D16) before anyone uses the system.*

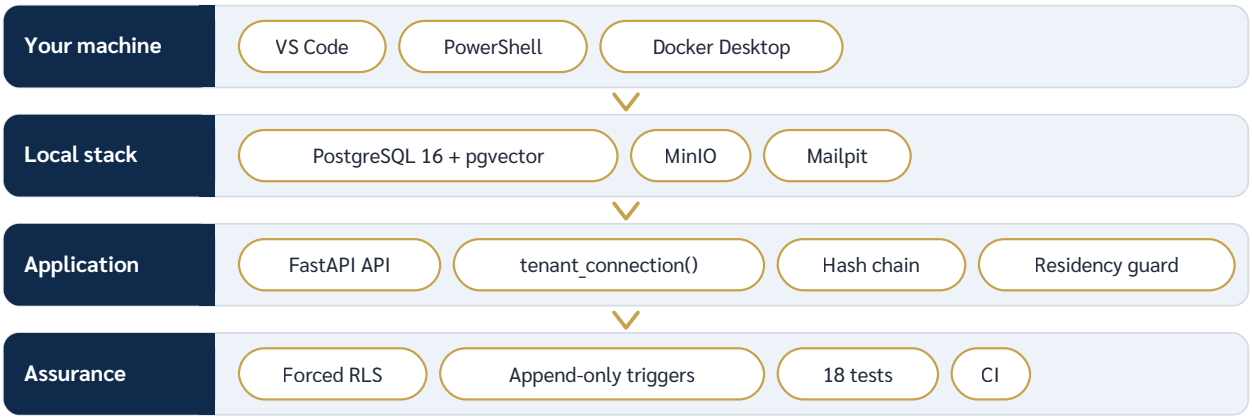
10	16/16	18	16
agent install tasks, T00 to T09	install checks to pass	automated tests	team tests before use

1 The pack

Part	What it is
TOOD-OS Full Master Blueprint v1.1	The specification. Section numbers in this guide refer to it
`tood-os-foundation-v1.1.zip`	The installable repository: local stack, database schema, tenant-safe access, provenance, AI residency guard, tests, CI, agent instructions
This guide	Install steps, agent protocol, acceptance checks, team test plan

The foundation is the base the rest of the platform is built on. It is not the finished platform: see section 10 for what remains.

2 What gets installed



Component	Local version	Purpose
PostgreSQL 16 with pgvector	Docker image `pgvector/pgvector:pg16`	Single database, forced row-level security, vector and full-text search
MinIO	Docker	S3-compatible object storage for uploads
Mailpit	Docker	Catches outgoing email locally
API	Python 3.12, FastAPI	Health endpoints, tenant context, hash chain, residency guard
Database schema	21 tables, 2 migrations	Blueprint Appendix A, with `app_user` and `auth_user` roles
Tests	pytest	18 tests covering isolation, append-only records, hash chain, residency
CI	GitHub Actions	Lint, migrations, verification, tests, secret scan, web build

3 Prerequisites

Need	Minimum
Git	Any current version
Python	3.12
Node.js	20 (for the web application step)
Docker Desktop	With Docker Compose v2, running
VS Code	VS Code offers to install the recommended extensions listed in `.vscode/extensions.json`
Windows	PowerShell 5.1 or later. If scripts are blocked: `Set-ExecutionPolicy -Scope Process Bypass`

4 Folder structure

```
tood-os/
├── AGENTS.md                rules every AI agent must read first
├── README.md
├── docker-compose.yml       postgres+pgvector, minio, mailpit
├── .env.example             local development values only
├── .vscode/                 extensions, settings, tasks, debug launch
├── .github/workflows/ci.yml
├── apps/
│   ├── api/                 FastAPI modular monolith
│   │   ├── app/core/        config, tenant_connection()
│   │   ├── app/ai_gateway/  residency guard
│   │   ├── app/provenance/  hash chain, ai_retained_ratio
│   │   └── tests/           18 tests
│   └── web/                 Next.js, created in task T05
├── packages/
│   ├── ui/                  design system
│   └── api-client/          generated OpenAPI client (task T06)
├── workers/                 job handlers
├── prompts/                 versioned prompts
├── rules/                   versioned compliance rules
├── evals/                   golden_set/, results/, thresholds.json
├── migrations/              0001_schema.sql, 0002_security.sql
├── infra/terraform/
├── docs/                    agent-tasks.md, install-report.md, test-plan.md, adr/
├── runbooks/
├── security/                threat-model.md
└── scripts/                 bootstrap, migrate, verify_install,
                             check_gates, export_openapi
```

5 Install in VS Code

1. Unzip `tood-os-foundation-v1.1.zip` and open the `tood-os` folder in VS Code. Accept the recommended extensions.
2. Open a terminal in VS Code and run the bootstrap for your system:

System	Command
Windows PowerShell	`./scripts/bootstrap.ps1`
macOS, Linux, WSL	`./scripts/bootstrap.sh`

3. The script creates the virtual environment, installs dependencies, copies `.env.example` to `.env`, starts the Docker services, applies the migrations and runs the verification. Success ends with `16/16 checks passed`.

4. Everyday commands are also in the VS Code task list (Terminal → Run Task): Stack up, DB migrate, Verify install, API run, Test api.

> **VS CODE USER SETTINGS (OPTIONAL)**
>
> The pack includes `settings.user.json` for your own machine: Copilot enabled except for `.env` files, telemetry off, PowerShell as the default terminal. Open it with Ctrl+Shift+P, "Preferences: Open User Settings (JSON)", and paste its contents. The project's own settings and recommended extensions (including GitHub Copilot, Copilot Chat and Claude Code) are already in `.vscode/` and need no action.

6 Agent execution protocol

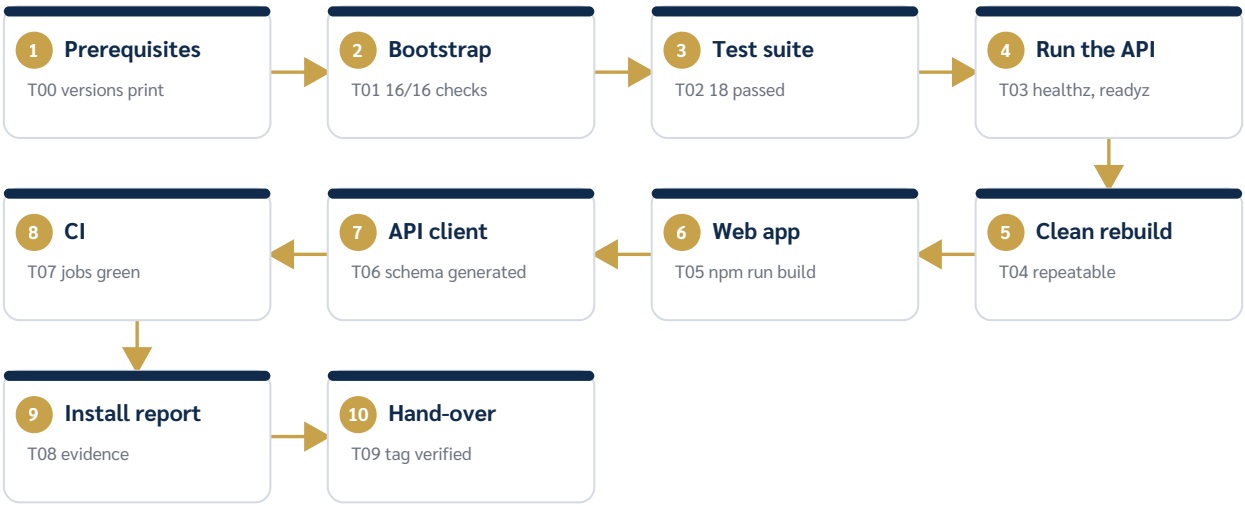
Give the agent this instruction, unchanged:

```
Open AGENTS.md and read it completely. Then execute docs/agent-tasks.md in order, T00 to
T09.
After each task run its acceptance check and record the result in docs/install-report.md.
Stop at the first failed check and report it. Do not edit a check to make it pass.
Work on a branch and open a pull request. Never push to main.
```

Rules the agent follows (full text in `AGENTS.md`):

- No production credentials, ever. Local and staging only.
- Every AI model call goes through the AI Gateway. No direct provider calls.
- Never weaken verbatim and claim verification, the export gate, provenance and hash chain, the Model Admission Test, forced RLS, append-only triggers, or the rule against automatic bid submission.
- Never use plain `SET` for tenant context. Use `tenant_connection()`.
- Never edit an applied migration. Add a new numbered file.
- Never commit secrets, `.env`, customer data or tender documents.
- Commit messages end with `Agent: <tool and version>`.
- A human must review authentication, RLS and migrations, billing, the AI Gateway, deletion, provenance and `security/` before merge.

7 Installation tasks



Task	Goal	Acceptance check
T00 Prerequisites	Machine can run the stack	Versions of git, Python (3.12+), Node (20+), Docker and Compose print
T01 Bootstrap	Environment, stack and database ready	Output ends `16/16 checks passed`
T02 Test suite	Foundation behaves as specified	`18 passed`, 0 failed, **0 skipped** (a skip means the database was unreachable)
T03 Run the API	API serves requests	`/healthz` returns `{"status":"ok"}`, `/readyz` returns `{"status":"ready"}`, `/docs` loads
T04 Clean rebuild	Install is repeatable	`docker compose down -v`, then T01 and T02 pass again. Running migrate twice prints `skip` the second time
T05 Web application	Next.js app created	`npm run build` succeeds in `apps/web`
T06 API client	Typed client generated	`openapi.json` and `schema.d.ts` exist and the web build still passes
T07 CI	Pipeline green	Every job in `ci.yml` passes on a pull request
T08 Install report	Evidence recorded	Every row of `docs/install-report.md` has a result and date
T09 Hand-over	Reviewed and tagged	A human merges the report pull request and tags `install-verified`

T05 and T06 commands:

```
npx create-next-app@latest apps/web --typescript --tailwind --eslint --app --use-npm --no-src-dir --import-alias "@/*"
cd apps/web && npx shadcn@latest init
python scripts/export_openapi.py
npx openapi-typescript packages/api-client/openapi.json -o packages/api-client/schema.d.ts
```

8 What success looks like

`python scripts/verify_install.py` checks sixteen things: Python version, `.env`, database connection, the three extensions (vector, pg_trgm, pgcrypto), all 21 tables, forced row-level security on all 18 tenant and identity tables, that `app_user` and `auth_user` are neither superuser nor `BYPASSRLS`, the three append-only triggers, default-deny tenant isolation, and the API health and readiness endpoints. Any `FAIL` line names what is wrong.

> **VERIFICATION STATUS**
>
> Before release of this pack, the following were run on Linux with PostgreSQL 16.15 and pgvector 0.6.0: both migrations (applied twice, second run skipped), the 18 tests (all passed), `verify_install.py` (16/16), and the linter.
>
> **Not run in that environment, so the development team must confirm them:** Docker Compose, the PowerShell bootstrap on Windows, the GitHub Actions run, and the Next.js steps T05 and T06.

9 Development team test plan

Run before any use. Each item is also in `docs/test-plan.md`.

ID	Area	How to run	Pass criteria	Blueprint
D01	Installation	`python scripts/verify_install.py`	16/16 checks pass	24, 30
D02	Migrations	Run `python scripts/migrate.py` twice	Second run skips every file	21
D03	Tenant isolation	`pytest tests/test_tenant_isolation.py`	6 pass	19, 22
D04	Default deny	No tenant context	Zero rows returned	19
D05	Cross-tenant write	Insert with another organisation's ID	Rejected by RLS	22

ID	Area	How to run	Pass criteria	Blueprint
D06	Append-only	`pytest tests/test_append_only.py`	App role cannot update or delete; trigger blocks the admin role	18
D07	Hash chain	`pytest tests/test_provenance_hash.py`	Tampering and removal detected at the right revision	18
D08	Residency guard	`pytest tests/test_residency_guard.py`	Non UK/EU regions rejected	12
D09	Role privileges	`app_user`, `auth_user`	Not superuser, no `BYPASSRLS`	19
D10	CI	Open a pull request	All jobs green	25
D11	Secrets	Gitleaks job in CI	No findings	25
D12	Web build	`npm run build` in `apps/web`	Succeeds	7
D13	API contract	`python scripts/export_openapi.py`	Client generated, web still builds	20
D14	Evaluation gates	`python scripts/check_gates.py`	Fails with no results; passes only when every threshold is met	28
D15	Clean machine	Fresh clone, bootstrap, on Windows and on macOS or Linux	Passes on both	30
D16	Rollback	`docker compose down -v`, bootstrap again	Passes	27

> ✓ **SIGN-OFF**
>
> All sixteen pass, results recorded, tester named and dated. Until then the system is not used with real data.

10 After installation: what remains to build

The foundation implements stage 1 of the blueprint's delivery plan (section 30) in part. The rest is built by the same agent workflow, one pull request per item, each with its tests and evaluation result.

Stage	Build items	Blueprint
1. Foundation (remaining)	Authentication, organisations and RBAC, file quarantine, safe parsing, document graph	13, 19
2. Intelligence	Requirement extraction with verbatim check, compliance matrix, evidence library and retrieval, drafting with claim verification, AI Gateway with model cards and admission tests	12, 14–17, 28
3. Assurance and collaboration	Provenance marks in the editor, daily anchoring, AI Use Record, export manifests, deletion, tasks and comments, clarification diff, billing, public API, SSO	18, 20, 27
4. Hardening	Security, accessibility, performance and recovery tests, external reviews	28, 34
5. Controlled release	Human release approval, controlled customer cohort	30, 35

11 Troubleshooting

Symptom	Cause and fix
`docker compose` **fails or hangs**	Docker Desktop is not running. Start it and retry
Port 5432 already in use	Another Postgres is running. Stop it, or change the port in `docker-compose.yml` and `.env`
`extension "vector" is not available`	The database is not the pgvector image. Use `pgvector/pgvector:pg16`
Tests are skipped	The database is unreachable. Fix the connection, then rerun. A skip is a failure
`password authentication failed for user app_user`	Run `python scripts/migrate.py` again. It resets the role passwords from `.env`
PowerShell blocks the script	`Set-ExecutionPolicy -Scope Process Bypass`, then rerun
`.env` **missing**	Copy `.env.example` to `.env`
Line-ending warnings on Windows	Keep `core.autocrlf=false`. `.editorconfig` sets LF
`create-next-app` **refuses the folder**	Delete `apps/web/.gitkeep` first

12 Hand-over checklist

- Install report complete, every task PASS, with dates and agent versions.
- Tag `install-verified` on a human-merged commit.
- Test plan D01–D16 signed off by the development team.
- No `.env`, secrets or real tender documents in the repository.
- Blueprint release checklist (section 35) opened and owned by a named person.

Contact

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